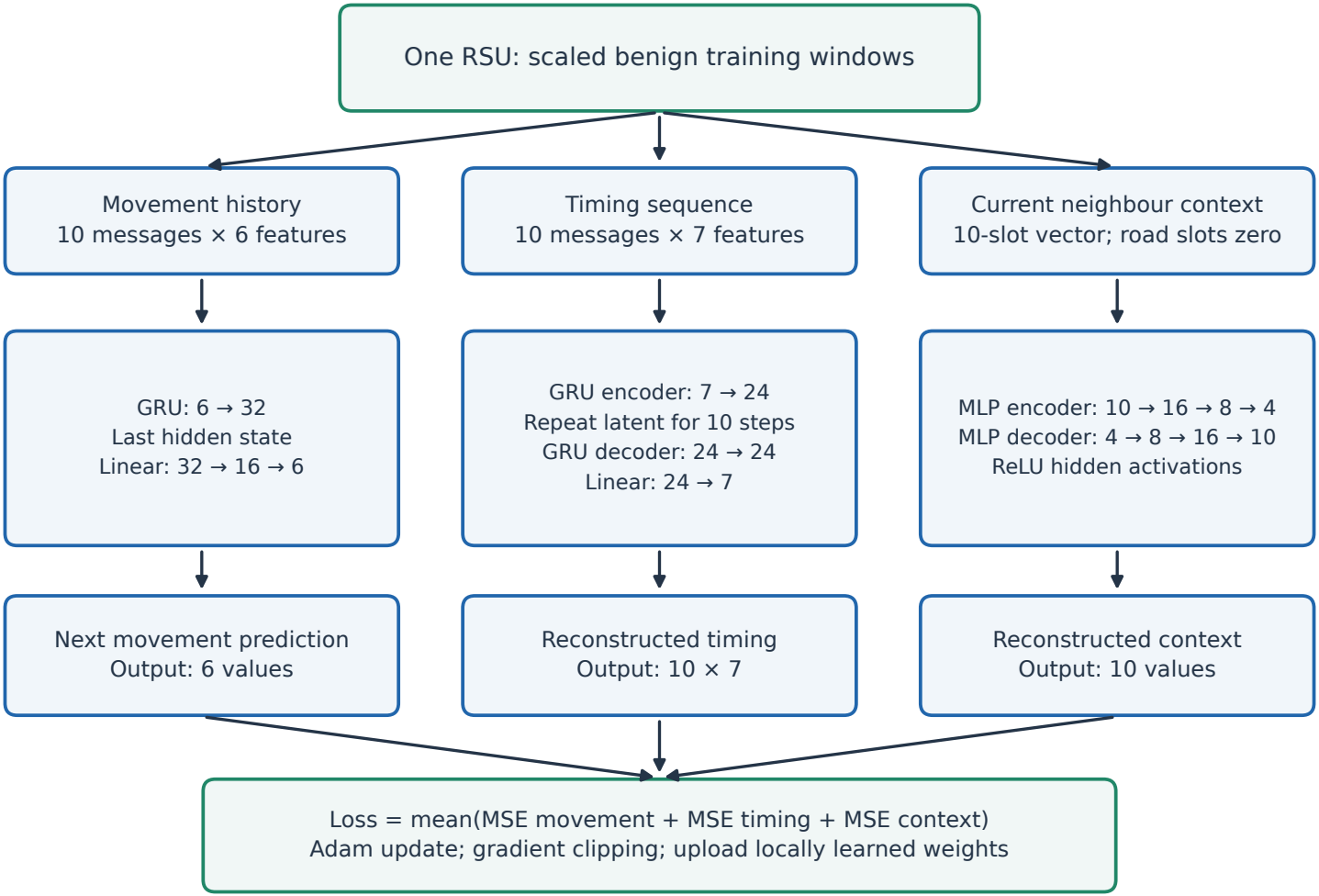
One RSU: scaled benign training windows
Movement history
10 messages × 6 features
Timing sequence
10 messages × 7 features
Current neighbour context
10-slot vector; road slots zero
GRU: 6 → 32
Last hidden state
Linear: 32 → 16 → 6
GRU encoder: 7 → 24
Repeat latent for 10 steps
GRU decoder: 24 → 24
Linear: 24 → 7
MLP encoder: 10 → 16 → 8 → 4
MLP decoder: 4 → 8 → 16 → 10
ReLU hidden activations
Next movement prediction
Output: 6 values
Reconstructed timing
Output: 10 × 7
Reconstructed context
Output: 10 values
Loss = mean(MSE movement + MSE timing + MSE context)
Adam update; gradient clipping; upload locally learned weights